

The Role of Artificial Intelligence for Cyber Attack Prediction: A systematic Literature Review

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Abstract

This systematic literature review examines how Artificial Intelligence (AI) is transforming its contribution to cyber attack prediction, intrusion detection, and phishing management in IoT, smart grids, and business networks. Consolidating trends, methods in the form of Machine Learning (ML) and Deep Learning (DL), datasets, and AI-based cybersecurity system performance results of PRISMA-guided 26 foundation studies spanning 2022–2025 were well explored. The findings indicate that hybrid and ensemble models all achieve better detection with more than 98% scores for the majority of cases, particularly when

they are tailored to specific deployment contexts such as resource-limited environments or real-time systems. Nevertheless, no model performs best on every evaluation metric or application domain. Explainable AI (XAI) techniques such as SHAP, LIME, and RuleFit seemed to enhance model transparency and analyst trust at the expense of performance or computational cost. Some of the open challenges that this review reveals include over-reliance on a small set of benchmark datasets (e.g., NSL-KDD, CICIDS2017, WSN-DS, UNSW-NB15 and BoT-IoT), minimal discussion about adversarial robustness, scalability problems, and absence of real-time validations. In the African context, the study underscores that these limitations are amplified by outdated infrastructure, lack of localized cybersecurity datasets, and limited AI deployment capacity. Finally, this work concludes by commenting on outstanding gaps in the research and future directions toward developing trustworthy, efficient, and explainable AI solutions suitable for Africa's evolving digital ecosystems.

Keywords: Cybersecurity, Artificial Intelligence, Deep Learning, GenAI, Cyber Attack.

1 Introduction

With fast technological evolution and increasing interconnectivity in our society, demand for security solutions and mitigation strategies will be increasingly important. Technological innovation simplifies life and makes it more convenient for all elements but on existing some challenges. One of the significant challenges is the fast development of cybersecurity attacks with technological innovation. With development in technology and companies shifting increasingly towards online platforms, the cyber-attacks have widened their range. These attacks could inflict irreparable damage upon organizations and individuals alike, which could lead to loss of money, bruised reputations, and even national security threats. It is therefore important that governments, businesses, and people all prioritize highest order to cybersecurity steps to secure their respective interests [1]. Taking all these into perspective, cybersecurity has grown much more crucial for professionals and researchers. It refers to an extensive range of elements such as tools, techniques, policies, security measures, guidelines, risk-mitigation techniques, training, best practices, and emerging technologies. These elements together strive to secure cyberspace and user assets [2]. Cybersecurity describes mechanisms that secure systems from threats and vulnerabilities to ensure the effective delivery of correct services to users. As a result of immediate escalation of data volumes, security has now been an imperative concern in cybersecurity. Based on studies, there is no doubt that cybercrime has caused harm to numerous organizations, companies, and individuals over the past few years [3–5]. As cyber threats emerge with increasing complexity and frequency, conventional cybersecurity methods fall short in detecting and reacting to emerging attack patterns [6]. Computer system cyber defense has become increasingly difficult. It is necessary to design more effective and effective cybersecurity practices to prevent cyberattacks. In response to this, Artificial Intelligence (AI) has emerged as a powerful tool to enhance cybersecurity

defenses. By harnessing AI capabilities like machine learning (ML), deep learning (DL) and Generative AI, cybersecurity appliances can now detect, analyze, and respond to cyber threats with unprecedented speed and accuracy [7–9]. Over the past decade, AI has been incorporated into different areas of cybersecurity, from intrusion detection systems (IDS) and malware detection to phishing detection and threat intelligence [10–12]. Such AI-powered tools have demonstrated significant potential in enhancing cyber threat detection, response, and automating complex security processes. However, the use of AI in cybersecurity also brings new challenges such as the need for quality training data, the risk of adversarial attacks against AI models, and ethical concerns with regard to data privacy and bias. To gain a view of the current situation and future directions of AI in cybersecurity, we conducted a systematic review of the literature over the past decade (2020-2025). This review is intended to trace the most crucial trends, developments, and challenges in the use of AI in cybersecurity.

2 Related Work

Abdullahi et al. (2022) [13] conducted a systematic review of the use of artificial intelligence (AI) methods to detect cybersecurity attacks on Internet of Things (IoT) networks. They considered 80 papers that were published during the period 2016-2021 and mentioned machine learning (ML) and deep learning (DL) methods for intrusion detection and anomaly detection in IoT networks. The authors categorized prior research by AI classifiers (e.g., Support Vector Machine (SVM), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN)), datasets used (e.g., NSL-KDD, UNSW-NB15, CICIDS2017, and IoT-related traffic), and types of attacks they identify (e.g., DoS, DDoS, botnets, ransomware, and reconnaissance). One of the strongest revelations made by Abdullahi et al.’s survey was that SVM and RF were the most commonly used ML methods due to their very high detection capability and low memory usage, and DL methods such as RNN and CNN performed best in handling high-dimensional and temporal network traffic. Hybrid AI models and fog computing-based IDS also emerged as future research directions. This review outlined a road map for AI on detecting cyber-attacks, overlaying ML/DL algorithms on anticipated types of attacks (CNN for DoS/DDoS, LSTM for sequential traffic anomaly). However, barely even mentioned detection algorithms and performance metrics the review did not delve into explainability, adversarial robustness, nor privacy-preserving learning methods, which are becoming more central to safe and trustworthy AI in cybersecurity.

Mendes and Rios (2023) [14] conducted a systematic review by focusing on Explainable Artificial Intelligence (XAI) techniques applied to cybersecurity. The main goal of this systematic review is to identify which XAI methods are currently being applied within cybersecurity systems, along with their areas of application that can benefit the most from explainable models. The review employed a systematic method on Scopus, ACM Digital Library, and IEEE Xplore and resulted in 42 unique references and 21 original studies after applying the inclusion criteria. The included studies covered a wide range of issues in cybersecurity including intrusion detection, malware detection,

phishing detection, anomaly detection, website fingerprinting, and security of mobile applications. According to the review, the top-rated post-hoc explanation techniques were SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME), followed by domain-specific techniques such as LEMNA and FAIXID. The majority of the research used them to improve analysts' confidence in AI-driven security systems, feature importance explanation, and model debugging. Additionally, visual analytics approaches and hybrid methods that incorporate domain knowledge infusion along with machine learning were considered new methodologies to enhance interpretability without a significant sacrifice in model performance. The review adequately highlighted some of the drawbacks of current research like a non-standard metric to measure explainability, compromise in performance in case of using intrinsically interpretable models, and minimal real-world evaluation of XAI-driven cybersecurity measures. It also mentioned that majority of the research was focused on additive local explanations of individual predictions and never evaluated the impact of explainability on decision-making accuracy or analyst effort.

Giarimpampa et al. (2025) [15] conducted a broad systematic literature review of Artificial Intelligence (AI) in Security Operations Centers (SOCs) and reviewed 189 articles between 2017-2023. They cross-mapped AI-enabled SOC solutions with NIST Cybersecurity Framework functions (Identify, Protect, Detect, Respond, and Recover) to analyze the degree to which AI is enhancing security operations. The assessment identified threat detection (Detect) as the most frequent function of AI tools utilized at 65% of the studies, and then comparatively lower studies on Respond and Recover functions. Among the reasons the authors cited for the adoption of AI in SOCs included improving threat detection accuracy, minimizing analysts' cognitive burden, facilitating real-time incident response, and giving scalability in responding to massive-scale cyber incidents. The authors also noted greater use of machine learning algorithms such as Random Forest, Support Vector Machines, and deep learning algorithms such as CNNs and LSTMs. But they also reported not enough use of explainable AI (XAI) methods, with 88% of research using black-box models and thereby losing transparency and analyst trust in AI-driven conclusions.

Faraji et al. (2024) [16] conducted a systematic review of artificial intelligence (AI) adoption to enhance cybersecurity in financial transactions. They meta-analyzed 63 qualified primary studies out of 800 qualified studies using PRISMA evidence synthesis guidelines on AI applications in fraud detection, anti-money laundering (AML), anomaly detection, identity-based attack prevention, and transaction monitoring. The study put the merging of machine learning (ML) and deep learning (DL) technologies in the spotlight and highlighted the real-time anomaly detection, behavior analysis, and automated incident response capability of the technologies. The study made use of AI's potential to reduce false positives and its contribution to predictive risk modeling in banking and finance networks. Traditional AI approaches—reinforcement learning for adaptive security and neural network-based threat detection—were identified as game-changers. Despite this, the survey also identified key gaps: (1) lack of explainable AI (XAI) methods for improved model interpretability and analyst confidence, (2) lack of empirical evidence of application to real-world financial systems, and (3) lack of consideration of computational efficiency and cost-benefit trade-offs

of AI adoption. Other concerns, such as quantum computing threats and outdated financial security paradigms, were argued to be potential weaknesses that should be addressed in future work. Recent studies reveal that Africa’s cybersecurity landscape faces distinctive challenges linked to its rapid digital transformation. Research such as Omollo and Maharaj (2023), Eltahir and Ahmed (2023), and Ndiaye et al. (2024) highlight that weak institutional frameworks, limited technical capacity, and inadequate awareness contribute to rising incidents of ransomware, phishing, and data breaches. Moreover, the reliance on outdated security tools and the absence of datasets reflecting African network traffic hinder the development of context-aware AI systems. These findings emphasize the urgent need for lightweight, explainable, and locally adaptable AI-driven cybersecurity solutions designed for Africa’s low-resource environments and growing digital infrastructure.

3 Research Methodology

This section describes the review methodology utilized to conduct a systematic review to investigate the role of Artificial Intelligence in cyber-attack prediction or intrusion detection. The review is conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) to ensure a transparent and reproducible study selection process. Specific search questions in Table 4, sources of databases in Table 4, and search strategies are defined in order to extract relevant studies in a systematic manner as illustrated in Table 4. Inclusion, exclusion, and screening criteria presented in Table 4 are defined in order to maximize high-quality research contribution selection. The review structure is designed to explore trends, algorithms, datasets, and evaluation metrics relevant to cyber-attack prediction or IDs.

3.1. Research Question

A systematic literature review (SLR) based on a research query means a systematic approach to locate, analyze, and synthesize the research evidence in response to the query given. In our systematic review here, we have come up with four main research queries, each seeking to address a specific issue on the role of Ai for cyber-security applications. These questions have been made unambiguous, concise, and in relation to the review purposes such that there is a systematic process of data analysis and collection. The research questions and corresponding objectives are provided in Table 3

3.2. The Search Databases and Search Queries

To obtain a comprehensive review of relevant studies, we conducted a literature search in primary academic databases, namely Google Scholar, IEEE Xplore, ScienceDirect, and Springer. The search strategy was designed to yield studies focused on deep learning (DL) and machine learning (ML) algorithms utilized for cyber-attack prediction, IDs and phishing detection and explainable techniques used in both machine learning and deep learning for cybersecurity applications. We designed focused search strings

Table 1 The summary of related SLR studies with their strengths and weaknesses

Author	Strength	Weakness
Abdullahi et al. (2022) [13]	- Reviewed 80 empirical studies (2016–2021) across DL/ML approaches. - Classified studies based on algorithms (ML, DL), IDS types, attack types. - Discussed commonly used AI algorithms like SVM, RF, CNN, RNN. - Presented a comprehensive discussion of datasets, machine learning algorithms and performance metrics.	- Limited on studies of 2016–2021, possibly missing relevant recent works. - Dependency on outdated datasets such as KDD99 and NSL-KDD. - Only accuracy metrics emphasized in performance comparison, omitting deeper statistical analysis
Mendes and Rios (2023) [14]	- The first review to address XAI application to cybersecurity. - Enumerates the most widely employed XAI techniques: SHAP, LIME, and a few recent ones LEMNA, SDL - classify explainability as intrinsic vs post-hoc, and local vs global..	- Limited to articles present in three databases (Scopus, IEEE, ACM), potentially excluding relevant work elsewhere. - Most articles are focused on Intrusion Detection, with the other being less prominent.
Faraji et al. (2024) [16]	- Provides a comprehensive SLR of 800 initially extracted papers, brought down to 63 core studies - Directs researchers attention to Ai based cybersecurity for financial application. - Categorizes modes of attacks and AI-based preventive strategies.. - Presents functions of AI in identifying fraud, predicting threats, and incident response	- Lacks strict adherence to systematic standards(PRISMA guidelines). - No comparison of performance on various state of art AI models. - Overemphasis on theoretical discussion with little reference to empirical studies
Giarimpampa et al. (2025) [15]	- Systematic review of 189 studies integrating AI in SOC's across the NIST Cybersecurity Framework. 6 - presented limited explainability integration in in previous AI based SOC's researches. - Present commonly used datasets and evaluates their relevance. - Adhere PRISMA protocol	- Overemphasis on Detect function; Respond and particularly Recover are underexplored. - Only 11.6% of reviewed studies provide explainability information; 88.4% rely on black-box models.

Table 2 Summary of the systematic review methodology.

Category	Details
Search Strategy	Relevant articles were retrieved from databases like IEEE Xplore, Google Scholar, ScienceDirect, and Springer using keywords as shown in Table 4.
Inclusion Criteria	Peer-reviewed articles published between 2022 and 2025 focusing on ML, DL-based approaches, and explainable AI techniques for cyber-attack prediction, phishing detection, and IDS detection.
Exclusion Criteria	Studies lacking empirical evaluations or not addressing cyber-attack prediction, phishing detection, and IDS detection; articles published before 2022; PhD dissertations; scientific reports; and magazines.
Data Extraction	Key information on model architecture, datasets, performance, feature selection techniques, and explainability methods was extracted.

Table 3 Research Questions (RQs) and their Objectives.

RQ	RQ Statements	RQ Objectives
RQ1	What are the various AI techniques that are most effective for cyber-attack prediction?	To identify and compare ML and DL models used for accurate cyber-attack detection.
RQ2	What are the impacts of ensemble and hybrid models in cybersecurity?	To evaluate how combining models improves detection accuracy and robustness, and to highlight their shortcomings.
RQ3	What are the challenges and areas of improvement in previous approaches?	To identify knowledge gaps and recommend future research directions in the field.
RQ4	What are the widely used XAI techniques in previous research?	To identify and summarize the most frequently applied XAI techniques in cybersecurity studies.
RQ5	What are the major cybersecurity challenges in Africa?	To identify the major cybersecurity challenges in Africa.

specifically to include studies concerning ML and DL perspective like "machine learning" OR "deep learning" OR "generative artificial intelligence" OR "GenAI" OR "explainable AI" and domain specific searching terms like "cyber-attack prediction" OR "malware detection" OR "intrusion detection" OR "phishing detection". The search strings utilized in our search are presented in Table 4.

Table 4 Search queries in the selected databases.

Database	Search Query
Google Scholar	("cyber-attack prediction" OR "malware detection" OR "intrusion detection" OR "phishing detection") AND ("machine learning" OR "deep learning" OR "generative artificial intelligence" OR "GenAI" OR "explainable AI") AND ("network security" OR "IoT security" OR "cybersecurity")
Springer	("cyber-attack prediction" OR "malware detection" OR "intrusion detection" OR "phishing detection") AND ("machine learning" OR "deep learning" OR "generative artificial intelligence" OR "GenAI" OR "explainable AI") AND ("network security" OR "IoT security" OR "cybersecurity")
IEEE Xplore	("cyber-attack prediction" OR "malware detection" OR "intrusion detection" OR "phishing detection") AND ("machine learning" OR "deep learning" OR "generative artificial intelligence" OR "GenAI" OR "explainable AI") AND ("network security" OR "IoT security" OR "cybersecurity")
Elsevier	("cyber-attack prediction" OR "intrusion detection" OR "phishing detection") AND ("machine learning" OR "deep learning" OR "generative artificial intelligence" OR "explainable AI") AND ("network security" OR "IoT security")

Table 5 Selected research databases.

No.	Database	Search Link
1	Google Scholar	https://scholar.google.com/
2	Springer	https://link.springer.com/
3	IEEE Xplore	https://ieeexplore.ieee.org/
4	ScienceDirect	https://www.sciencedirect.com/

3.3. Inclusion and Exclusion Criteria

To select relevant and good-quality studies, inclusion and exclusion criteria were defined. The inclusion and exclusion criteria were used during screening to deselect poor-quality or irrelevant studies. Studies were selected based on relevance to cyber-attack prediction, IDs and phishing detection, methodological quality, and contribution to the literature as a whole. Additionally, the articles must be published in

reputable journals or top-tier conferences, have a publication date no older than 2022, and address at least one of the defined research questions. In addition to global sources, this review also examined African-based studies and policy reports [17–19] to interpret findings through a regional lens. The selection process paid particular attention to works highlighting AI applicability in low-resource African environments, emphasizing capacity, infrastructure readiness, and ethical AI adoption challenges. Table ?? lists the entire inclusion and exclusion criteria adopted in this review.

Table 6 Selection criteria.

No.	Inclusion	Exclusion	Description
I1	✓	–	Studies published between 2022–2025.
I2	✓	–	Peer-reviewed journal articles and conferences written in English with title and abstract.
I3	✓	–	Studies focused on IDS detection, cyber-attack prediction, explainability, or phishing detection.
I4	✓	–	Studies with sufficient methodology, experimental results, and performance metrics.
E1	–	×	Scientific reports, books, PhD dissertations, magazines, or studies written in other languages.
E2	–	×	Studies published before 2022.
E3	–	×	Studies not focused on IDS, phishing, or cybersecurity-related detection.
E4	–	×	Studies lacking explainability, ML/DL approaches, or methodological clarity.
E5	–	×	Studies with insufficient experimental results or missing performance metrics.

4 Machine Learning Approach

The majority of malware detection tools have traditionally been proposed on the basis of machine learning. In this subsection, we will present about some research works in the area of the proposed systems. BIRAHIM et al., (2025) [20] presents a robust Wireless Sensor Network (WSN) intrusion detection system (IDS) that is an integration of Particle Swarm Optimization (PSO) with Random Forest, Decision Tree, and K-Nearest Neighbors ensemble classifiers. The paper also addresses significant problems such as class imbalance, excessive false positives, and interpretability. SMOTE-Tomek is employed for data balancing and explainable AI techniques SHAP and LIME are

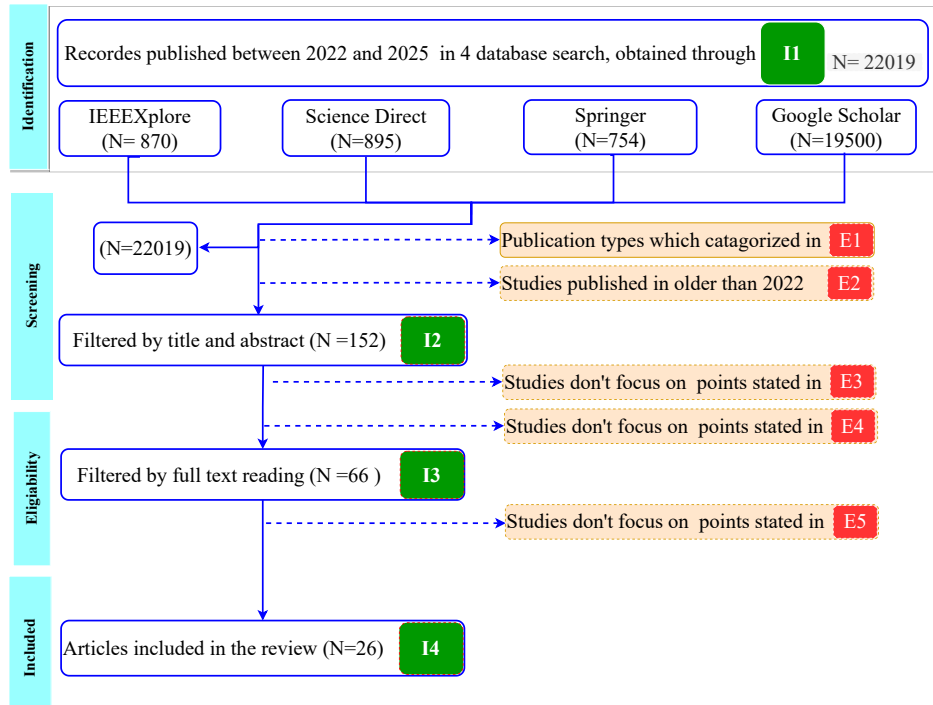


Fig. 1 Flow diagram of paper selection

utilized for local and global interpretability. Measured against the WSN-DS dataset, it achieves 99.73% accuracy with precision, recall, and F1-score of 99.72% all. A Streamlit web interface also enhances usability to a greater extent. Compared to current works like CNN and CatBoost-based models, it offers improved accuracy, explainability, and a well-balanced model performance and interpretability approach. Its computational complexity may limit real-time usage and its testing is limited to a single dataset. In

Table 7 Search Database and Filtering Results.

Selection criteria	Google Scholar	IEEE Xplore	Springer	ScienceDirect
Search by keyword	21300	9600	6137	2964
Filtering by I1 & I2 (without title and abstract)	19500	870	754	895
Filtering by I3 , title & abstract (I2)	34	41	42	35
Filtering by I4	5	13	6	2

general, this work advances a highly accurate, interpretable, and scalable IDS solution with practical relevance to WSN applications. Barnard et al., (2022) [21] presents a novel two-stage pipeline for robust network intrusion detection with supervised and unsupervised learning and explainability. In the first stage, an XGBoost classifier is used for binary attack detection in the NSL-KDD dataset, and SHAP explanations are used to identify feature importance. SHAP vectors are then inputted to a deep autoencoder in the second stage for classification between known and new (zero-day) attacks via anomaly detection. This new use of XAI outputs as inputs to an unsupervised model is a unique methodological innovation. Evaluation shows that while the supervised model alone had 79% accuracy with high precision (96%) but low recall (65%), and the anomaly detector had high recall (89%) but low precision (43%), their combination significantly outperformed state-of-the-art methods with 93% accuracy, 91% precision, and 97% recall. Despite the success, limitations include dependence on a single dataset and binary classification. Nevertheless, this study contributes to the field by demonstrating how XAI can be used to improve both detection and explainability in intrusion detection systems. A. ELSADIG et al., (2023) [22] addresses the growing vulnerability of Wireless Sensor Networks (WSNs) to Denial-of-Service (DoS) attacks, by proposing a light-weight detection model based on Decision Tree (DT) classifiers optimized by feature selection based on Gini. In this work, recent ML and DL-based DoS attack detection techniques and their shortcomings and computational inefficiencies in particular for resource-constrained environments like WSNs are reviewed in detail. One of the significant contributions is the enhancement of the WSN-DS dataset using feature selection, which removes the major portion of computational overhead but maintains classification accuracy. Comparative experiments with RF, KNN, and XGBoost confirm that the DT model achieves 99.5% accuracy consuming only 9.7%, 13%, and 2% of RF's, XGBoost's, and KNN's processing time. The lightweight nature of this method makes it extremely conducive for real WSN deployment. However, limitations are the application of a single, unbalanced dataset and failure to generalize to adversarial attacks or other classes of networks. Further research is necessary to test on several datasets and balance class distribution to improve robustness. Overall, this proposed an energy-efficient ML-based detection scheme specific to WSNs' computational constraints. The exponential rise in phishing attacks has become a pressing threat to internet security, exploiting both system vulnerabilities and human psychology. Anti-phishing blacklists are a major defense mechanism, they have the limitation of incomplete coverage and late updates and hence are prone to be evaded by advanced attackers. LI et al., (2024) [23] provides an extensive security analysis of blacklists used against anti-phishing and proposes two new cloaking attacks Feature-Driven Cloaking and Transport Layer Security (TLS)-Based Cloaking that take advantage of weaknesses in the automated detection system of anti-phishing entities (APEs). From real data and using machine learning approaches, the RF classifier was the most robust among all supervised classifiers that were experimented with a 100% success rate in distinguishing APEs from regular users and enabling attackers to bypass blacklist detection. The primary findings point to severe security flaws in well-known APEs like restricted infrastructure diversity, feature implementation inconsistency, and vulnerability to Web Real-Time Communication (WebRTC) Internet Protocol (IP) leaks.

These weaknesses contribute to the operational lifetime of phishing pages, which pose greater threats to users. The results point towards the need for APEs to implement stronger and adaptive protections and suggest countermeasures to enhance the resilience of the anti-phishing ecosystem.

Mohamed et al., (2023) [24] examines the performance of various machine learning models for intrusion detection on the UNSW-NB15 dataset, a standard dataset with mixed benign and malicious network traffic. It is comparing the performance of classical ML models (Decision Trees, SVM, Random Forest) and deep models and compares them on the measures of accuracy, precision, recall, and F1-score. It also highlights the significance of feature engineering for enhancing detection performance. Its principal contribution is its comprehensive comparison of different models and the impact of feature extraction strategies on intrusion detection performance. The piece is, however, limited in that it does not provide real-world deployment validation and lacks a new model or optimization strategy. It does not even address adversarial robustness and energy efficiency as central concerns for feasible IDS deployment. However, this work provides valuable comparative comments regarding ML-based IDS design and acts to guide the selection of suitable models and features for improving cybersecurity with the UNSW-NB15 dataset.

5 Deep Learning Approach

Deep learning (DL) is presently a cornerstone in modern intrusion detection systems (IDS) for foretelling and defending against cyber-attacks. Unlike the conventional machine learning models, deep learning can automatically extract complex hierarchical features from raw data, thus facilitating robust detection of known attacks and zero-day attacks [25, 26]. Modern surveys quote LSTM, CNN, and ensemble hybrid models trained on benchmark datasets such as UNSW-NB15 and CICIDS2017 as state-of-the-art to detect evasive threats and zero-day attacks. For instance, Xu et al. (2025) [27] provide an in-depth overview of DL-based IDS development, spanning from parsing logs to detecting attacks and demonstrating existing difficulties in terms of generalizability and real-time deployment. Ahmad Umar et al., (2025) [28] proposed an energy-efficient Intrusion Detection System (IDS) architecture for edge computing environments using an adapted Deep Neural Network with Knowledge Distillation and Quantization (DNN-KDQ). To overcome the limitation of traditional IDS on resource-constrained devices, the study proposes model compression and adaptive sampling methods brought to the CICIDS2017 dataset. Major contributions include the energy-aware feature extraction, drastic model size reduction (from 196.77 KB to 20.18 KB), and inference time acceleration of 0.07 ms per sample—with the high test accuracy of 99.43%. These results demonstrate the model’s viability for real-time anomaly detection in edge and IoT-based environments. However, the paper does not mention adversarial robustness and neither generalizability to other networks nor other network conditions. Furthermore, even though compression and distillation perform well, explainability of the resulting quantized model is left untouched.

Haozhe Zhang (2025) [29] suggests implementing a deep learning- and ensemble-based intelligent Intrusion Detection System (IDS) for Internet of Things (IoT)

Table 8 The contribution and limitation of various ML approaches for Intrusion Detection

Author (Year)	Key Contribution	Limitation
Birahim et al. (2025) [20]	Integrated PSO with ensemble of(Random Forest, Decision Tree, and K-Nearest Neighbors) classifiers with explainability techniques	High computational complexity unsuitable for real-time use; validation limited to one dataset
Barnard et al. (2022) [21]	presents a novel two-stage pipeline for robust network intrusion detection, Innovative use of explainable feature outputs as inputs to anomaly detector	Limited to binary classification and one dataset(NSL-KDD). The 100% accuracy claim may be dataset-specific and not generalizable.
A. Elsadig et al. (2023) [22]	High accuracy with low computational cost suitable for low-resource devices	Single unbalanced dataset; no adversarial robustness analysis
Li et al. (2024) [23]	Exposed critical weaknesses of anti-phishing entities and proposed countermeasures	Focus on attack strategies without new defensive mechanism
Mohamed et al. (2023) [24]	comparison of classical ML and deep models using performance metrics.	No novel model or optimization; lacks real-world validation

networks to address IoT-specific security challenges such as device heterogeneity and large-scale, diverse attacks. The authors created 21 individual neural network models and an ensemble learning model with the BoT-IoT dataset and compared their performance based on accuracy, precision, recall, and F1-score. The ensemble model outperformed the rest with 99.985% accuracy, 99.28% precision, 99.80% recall, and 99.54% F1-score—outperforming state-of-the-art methods in detection performance. The key contribution is demonstrating that fusion of deep models improves detection robustness to diverse threats, offering a scalable network-level security solution for IoT networks. However, limitations include the absence of testing on real-time or resource-constrained IoT settings, lack of explainability analysis, and no benchmarking on additional datasets.

Table 9 Summary of Intrusion Detection Studies using ML Approaches

Author (Year)	Problem Addressed	Model	Dataset	Performance
Birahim et al. (2025) [20]	WSN intrusion detection with high false positives, class imbalance, and interpretability issues	PSO with (Random Forest+DT+KNN) ensemble; SMOTE-Tomek for balancing; SHAP & LIME for explainability	WSN-DS	Accuracy:99.73%, Precision:99.72%, Recall: 99.72%, F1: 99.72%
Barnard et al. (2022) [21]	Robust and explainable intrusion detection including zero-day attacks	Two-stage pipeline: (1) XGBoost with SHAP feature importance; (2) Deep Autoencoder using SHAP outputs	NSL-KDD	Accuracy: 93%, Precision: 91%, Recall: 97%
A. Elsadig et al. (2023) [22]	DoS attack detection in resource-constrained WSN environments	Lightweight Decision Tree optimized by Gini-based feature selection	WSN-DS	Accuracy: 99.5%, computation reduced to 9.7% of RF, 13% of XGBoost, 2% of KNN
Li et al. (2024) [23]	Phishing attacks bypassing anti-phishing blacklists	Two new cloaking attacks with RF classifier for APE detection	Real phishing data	Accuracy: 100% success in bypassing detection
Mohamed et al. (2023) [24]	Comparison of ML models for intrusion detection	Decision Tree, SVM, Random Forest, and Neural Networks	UNSW-NB15	Neural Networks outperformed with 98.3% accuracy

Alsirhani et al., (2023) [30] presents a hybrid intrusion detection methodology optimized especially for smart grids, addressing the issue of high false alarms and low detection accuracy in traditional IDS solutions. The proposed framework is based on deep learning and feature engineering, utilizing a Deep Belief Network–Long Short-Term Memory (DBN-LSTM) model optimized by the African Vulture Optimization Algorithm (AVOA). The pipeline begins with Min–Max normalization and statistical and information-theoretic feature extraction (e.g., entropy, percentiles, correlation). Optimal feature selection is subsequently applied by AVOA to increase the efficiency of the model and reduce redundancy. The DBN-LSTM classifier finally identifies traffic as normal or malicious. Experimental evaluations indicate that the proposed AVOA-DBN-LSTM model exhibits better classification accuracy and robustness compared to existing methods and proves to be effective in smart grid cybersecurity. But the work is limited by not considering deployment analysis under real-time constraints and does not consider computation or energy efficiency—significant for grid-level applications. Nevertheless, the hybrid strategy offers strong direction towards the convergence of optimization and deep learning for reliable intrusion detection in smart energy infrastructures. Abiodun Ayantayo et al., (2023) [31] introduce three novel deep network architectures early-fusion, late-fusion, and late-ensemble that are designed to enhance multiclass classification accuracy in Network Intrusion Detection Systems (NIDS). The novelty here is the use of feature fusion mechanisms for enhancing the generalization capability of fully connected deep networks, particularly for handling heterogeneous input features. The models are evaluated on two benchmarking datasets, UNSW-NB15 and NSL-KDD, and showed to address common problems such as class imbalance and overfitting. Among them, late-fusion and late-ensemble models generalize more effectively, with significant performance on both training and validation sets. A novel contribution of this paper is the empirical evidence that feature fusion enhances learning of inter-feature relationships, which otherwise skew the model. However, the limitation exists in that there is little analysis of computational costs, real-time inference performance, and deployment feasibility under resource-limited settings.

6 Ensemble Learning Approach

The ENIDS method by Sayem et al., (2024) [32] proposed a stacking ensemble approach with deep learning for multi-class network intrusion detection with CNN, LSTM, and GRU as base learners and DNN as a meta-learner. Two recent and widely used datasets UNSW-NB15 and CICIDS-2017 are used in the paper with extensive pre-processing, RFECV for feature selection, and SMOTE-ENN for class imbalance handling. The ENIDS model outperforms individual DL models and current state-of-the-art methods with up to 99.6% accuracy and F1-score on CICIDS-2017 and 90.6% accuracy on UNSW-15. The model achieves this with robust ensemble architecture, feature fusion, and real-world attack classification. The model, however, has higher training/testing time, and lacks full explainability mechanisms, which could limit transparency and adoption in sensitive cybersecurity applications. Abdullah Alabdulatif (2025) [33] presents a hybrid ensemble intrusion detection model that targets enhanced detection accuracy and explainability for real-time cyber use. The model

Table 10 Summary of Intrusion Detection Studies DL Approaches

Author (Year)	Problem Addressed	Model	Dataset	Performance
Umar et al. (2025) [28]	Energy-efficient IDS for edge computing	DNN with Knowledge Distillation & Quantization (DNN-KDQ) + adaptive sampling	CICIDS2017	Accuracy: 99.43%, Model size: 20.18 KB (from 196.77 KB), Inference: 0.07 ms/sample
Zhang (2025) [29]	IoT-specific IDS challenges	Ensemble models (CNN-GRU(2)+CNN-LSTM(2) + DNN(2))	BoT-IoT	Accuracy: 99.985%, Precision: 99.28%, Recall: 99.80%, F1: 99.54%
Alsirhani et al. (2023) [30]	Smart grid IDS false alarms & low accuracy	Hybrid AVOA-DBN-LSTM with feature engineering	Smart grid dataset	Accuracy: 98.99%, Precision: 96.97, Recall: 96.97, F1_score: 96.97%
Ayantayo et al. (2023) [31]	Improving multiclass classification & generalization	late-ensemble model(feature fusion)	UNSW-NB15 & NSL-KDD	Accuracy: 86.81%, Precision: 86.86% & Recall: 86.80%

employs a deep learning-based Artificial Neural Network (ANN) and Support Vector Machine (SVM) as base learners and a Random Forest meta-classifier to fuse predictions. Recursive Feature Elimination (RFE) is employed for best feature selection to remove redundancy and improve computational efficiency. Explainable Artificial Intelligence (XAI) techniques, such as SHapley Additive Explanations (SHAP), are utilized to provide global and local explainability, solving the "black box" issue generally associated with AI-driven IDS solutions. Experimental evaluation on the NSL-KDD dataset represents improved performance (accuracy of 99.40%) and satisfactory precision, recall, and F1-scores versus standalone classifiers. The system also demonstrates its deployment feasibility with operation on Amazon EC2 with a web interface with Flask for real-time traffic analysis and inference times in sub-seconds. While the approach is better in terms of interpretability and accuracy, limitations are single-dataset testing and limited mention of scalability across large-volume enterprise networks. Nevertheless, it considerably advances explainable and operationally deployable AI-based IDS solutions for critical areas such as finance, healthcare, and industrial IoT. Rahman et al., (2025) [34] presents an optimized light-weight Convolutional Neural Network (CNN) model for Distributed Denial of Service (DDoS)

Table 11 The strength and limitations of Deep learning methods used for IDs

Author (Year)	Key Contribution	Limitation
Umar et al. (2025) [28]	Compressed and energy-efficient IDS with high accuracy for edge computing	No adversarial robustness or explainability analysis
Zhang (2025) [29]	Ensemble deep models improved IoT IDS detection performance	No real-time deployment, single dataset, no explainability
Alsirhani et al. (2023) [30]	Hybrid optimization + deep learning for smart grid IDS	No deployment or computational cost analysis
Ayantayo et al. (2023) [31]	Feature fusion improved generalization in IDS models	No computational or real-time deployment analysis

detection, which is optimized to be deployed on resource-constraint environments. It provides a new light-weight CNN architecture with reduced parameters and computational cost without compromising competitive performance. The introduced model is trained and validated on the CIC-DDoS2019 dataset and achieved 99.7% accuracy, 99.6% precision, and 99.5% recall, making it reliable for real-time intrusion detection. The work highlights a fantastic trade-off between model performance and computational expense. The primary advantages of the the presented article are low resource consumption, fast inference time, and good performance on different attack classes. Nevertheless, the paper misses great elaboration on model interpretability and generalizability across data sets, which are significant factors to bear in mind during deployment robustly.

Ammar Odeh and Anas Abu Taleb (2023) [35] presented a deep learning architecture for IoT network cyber-attack detection from the BoT-IoT dataset based on a hybrid CNN-GRU model. The model makes use of CNN to extract spatial features and GRU to learn temporal dependencies, thus successfully handling the problem of detecting multi-vector attacks in heterogeneous IoT settings. Experimental results indicate over 99% accuracy, high recall and precision, beating conventional ML baselines and other DL configurations like LSTM. Its key benefits include robust detection of low-frequency attacks, temporal-spatial feature fusion, and low false positive rate. The paper also emphasizes scalability and real-time usage and can be utilized in resource-constrained IoT scenarios. However, the work does not deal with model interpretability and transferability to out-of-sample data distributions, thus limiting its explainability and generalizability essential considerations for security-critical applications. Broadly speaking, this work is significant for SLRs on effective, efficient, and flexible DL-based IoT threat detection.

Musthafa et al., (2024) [36] proposed an optimized IoT intrusion detection technique through the combination of class balancing, feature selection, and ensemble machine learning algorithms. The study targets key issues in IDS performance like overfitting and irrelevant features and applies the solution to two benchmark datasets UNSW-NB15 and NSL-KDD. It evaluates two ensemble algorithms: SVM with bagging and LSTM with stacking, with ANOVA-based feature selection. The LSTM-stacking model outperforms the alternatives with accuracy as much as 99.77% and overfitting as low as 0.04%, as well as AUC values approaching 1.0, revealing good generalization. Its main strengths include dual-dataset validation, careful handling of class imbalance, and minimal overfitting. Its limitations include insufficient discussion of computational efficiency, real-time scalability, and explainability vital for real-world deployment in IoT environments.

A light-weight ensemble learning algorithm for network intrusion detection was proposed by [37] to address the growing security threats in today’s networks. The ensemble model used Gaussian Naive Bayes, Logistic Regression, and Decision Tree as the individual classifiers and Stochastic Gradient Descent (SGD) as the meta-classifier. It utilizes Chi-square feature selection to enhance the performance of the model and reduce dimensionality. It is experimented on three common datasets KDD Cup 1999, UNSW-NB15, and CIC-IDS2017 with both binary and multi-class classification capability. The new ensemble outperforms individual models in terms of accuracy and robustness, particularly in dealing with imbalanced datasets, a crucial problem in IDS implementations. Below are its strengths: computation efficiency, multi-dataset testing, and application of interpretable low-complexity models, which make it suitable for real-time and resource-constrained environments. However, the paper does not address model interpretability at the ensemble level, nor deployment settings, thereby limiting practical understanding.

7 Hybrid Learning Approach

S. HARIHARAN et al., (2025) [38] presents a better hybrid model for Network Intrusion Detection Systems (NIDS) that combines Sequence-to-Sequence (Seq2Seq) structure with Convolutional LSTM (ConvLSTM) cells to address limitations of detecting sophisticated cyber attacks with heavy temporal and spatial dependencies. The proposed model uses CNNs for extracting spatial features and LSTM networks to learn sequences, thus able to handle variable sequence lengths and long-range dependencies. Specifically, the model includes Explainable AI (XAI) on the LIME platform, which provides added transparency through the identification of spatial-temporal features that contribute to detection decisions. Evaluated on three benchmark datasets CIC-IDS 2017, CIC-ToN-IoT, and UNSW-NB15 the methodology illustrates higher accuracy and lower false positives than existing models. Its main contribution lies in both optimizing detection performance and interpretability simultaneously, offering a more reliable and effective IDS. However, the study does not address deployment constraints such as computational overhead, memory requirements, or latency, which are essential in real-time or edge settings. Despite this, it is a fundamental contribution to designing intelligent, interpretable, and high-performance NIDS architectures.

Table 12 Summary of Ensemble Learning Based IDS Approaches

Author (Year)	Problem Addressed	Model	Dataset	Performance
Sayem et al. (2024)[32]	Multi-class intrusion detection with class imbalance and improved accuracy	Stacking ensemble with CNN, LSTM, GRU as base learners, DNN as meta-learner; RFECV for feature selection	UNSW-NB15, CICIDS-2017	proposed ENIDS : Accuracy: 99.6%, F1_score: 99.6% (CICIDS-2017); Accuracy: 90.6% (UNSW-NB15)
Abdullah Alabdulatif (2025) [33]	Real-time intrusion detection with explainability	(ANN+SVM) as base learners; Random Forest as meta-classifier; RFE feature selection	NSL-KDD	Accuracy: 99.40%, precision: 99%, recall: 98.5% & F1_score: 98%
Rahman et al. (2025) [34]	Lightweight DDoS detection in resource-constrained environments	Optimized lightweight CNN with reduced parameters and computational cost	CIC-DDoS2019	Accuracy:99.7%, Precision: 99.6%, Recall: 99.5%
Odeh & Taleb (2023) [35]	Multi-vector IoT cyber-attack detection	Hybrid CNN-LSTM model	BoT-IoT	Accuracy: 99.734%, precision: 99.7%, Recall: 99.9% & F1_score: 99.8%
Musthafa et al. (2024) [36]	IDS performance improvement addressing overfitting and imbalance	LSTM stacking ensemble with ANOVA-based feature selection (compared with SVM + bagging)	UNSW-NB15, NSL-KDD	Accuracy: 99.73% & 99.62% on UNSW-NB15, NSL-KDD respectively
Thockchom, (2024) [37]	Lightweight ensemble for network intrusion detection	Gaussian NB, Logistic Regression, Decision Tree as base models, SGD as meta-classifier; Chi-square feature selection	KDD Cup 1999, UNSW-NB15, CIC-IDS2017	ensemble model outperformed with accuracy of 99.84 (KDD'99) and 93.88 (UNSW-NB15)

Table 13 Contribution and Limitations of Ensemble Learning Approaches

Author (Year)	Key Contribution	Limitation
Sayem et al. (2024) [32]	Robust stacking ensemble integrating multiple deep learners with advanced feature selection and imbalance handling	High training/testing time, lacks comprehensive explainability
Alabdulatif (2025) [33]	Explainable ensemble IDS with real-time deployment capability	Tested only on one dataset, scalability for large networks untested
Rahman et al. (2025) [34]	Lightweight CNN balancing accuracy and computational efficiency	No interpretability analysis, single dataset focus
Odeh & Taleb (2023) [35]]	Hybrid CNN-GRU detecting complex IoT multi-vector attacks	Lacks interpretability and transferability evaluation
Musthafa et al. (2024) [36]	LSTM stacking ensemble with dual-dataset validation and minimal overfitting	Missing computational efficiency and deployment feasibility analysis
Thockchom, (2024) [37]	Low-complexity ensemble with multi-dataset validation and good robustness	Ensemble-level interpretability not addressed, deployment not analyzed

ALSUBAEI et al., (2024) [39] presents a state-of-the-art deep learning architecture for real-time phishing attack detection, with emphasis on detection performance and computational cost efficiency for digital forensics and wireless communication. The method synergizes an ensemble feature engineering pipeline composed of autoencoders and ResNet (EARN) for strong feature extraction with a ResNeXt network integrated with Gated Recurrent Units (GRU) as the backbone classifier. To address data skewness, the Synthetic Minority Over-sampling Technique (SMOTE) is employed, and hyperparameters are tuned by the researcher with the Jaya optimization algorithm. Experimental results on real-world phishing datasets show remarkable improvement in performance, including up to 98% accuracy and outperforming state-of-the-art models by 11%–19%, with low false positive and false negative rates. Computational effectiveness is also ensured, with average running times being 36.99s and minimal variance. The key contribution is that this method unifies optimized feature extraction, data balancing, and deep learning in one solution, with high applicability to organizational cybersecurity as well as digital forensics. However, the possibility of deployment on resource-constrained devices and adversarial robustness or explainability are not covered in the paper. Despite these shortcomings, the contribution presents a fine and efficient AI-driven phishing detection model for the enterprise domain.

SHAHID et al., (2024) [40] introduces an Intrusion Detection System (IDS) for Internet of Things (IoT) networks operating on the Routing Protocol for Low-Power

and Lossy Networks (RPL) that detects four key attacks: Black Hole, Flooding, DODAG Version Number, and Reduced Rank. Utilizing the recently crafted ROUT-4-2023 dataset, the authors compare network traffic characteristics using statistical information graphs and evaluate several machine learning and deep learning models. Results are 99% accurate for the Random Forest classifier, and an optimal training of a deep learning architecture founded on the Transformer achieves a 97% F1-score (16.8 minutes for five iterations). Key contributions are presenting an extensive IoT RPL dataset with multiple attack types and demonstrating Transformer architectures for IoT security operations. But the work does not address resource constraints of IoT devices for real-time deployment, model decision interpretability, or adversarial robustness. Despite such limitations, the study offers an effective and data-intensive approach to further advancing IDS research in resource-constrained IoT environments.

Abuali et al., (2023) [41] proposes a hybrid IDS based on the fusion of Support Vector Machine (SVM) and deep learning techniques for multiclass intrusion detection in social media networks. The researchers use the CSE-CIC-IDS 2018 dataset and conduct rigorous preprocessing, with model testing on 100,000 data samples. The proposed system has very high performance with 100% accuracy, recall, precision, specificity, and F-score, and eliminates false positives (0% false positive rate). The study demonstrates the potential of combining SVM and deep learning to process massive and complex network traffic data in dynamic social media. However, the perfect performance metrics point toward possible overfitting because the results are from a single dataset and limited experimental scope. The study also neglects to examine computational efficiency, scalability to real-time environments, and model interpretability. Despite these limitations, this research sheds light on the tremendous prospects of hybrid AI-based IDS solutions in revealing nascent cyber threats on social media networks.

Zahid and Bharati (2025) [42] proposes a hybrid deep model based on the combination of Convolutional Neural Networks (CNN) for high-dimensional feature learning and Bidirectional Long Short-Term Memory (BiLSTM) for temporal dependency modeling, to improve cyberattack detection in IoT networks. The suggested model integrates the Least Absolute Shrinkage and Selection Operator (LASSO) as a feature selection process to obtain computational efficiency and uses the Synthetic Minority Over-sampling Technique (SMOTE) to manage class imbalance. Examined on three datasets—KDDCup99, NSL-KDD, and CIC-IDS.2017—the model yields high accuracy (99.9%, 99.8%, and 98.0%, respectively) and stable performance for sophisticated types of attacks such as DoS, DDoS, and malware. Compared to the state-of-the-art techniques, the hybrid approach achieves improved precision, recall, and F1 score, showing promising performance for real-time IoT deployment. While the approach prioritizes detection precision, it does not prioritize deployment feasibility or model interpretability for low-resource IoT devices. Despite these limitations, the work presents a scalable and robust hybrid deep learning approach to enhance the security of IoT networks against a variety of ever-changing threats.

Table 14 Summary of Hybrid Learning Approach

Author (Year)	Problem Addressed	Model	Dataset	Performance
Hariharan et al. (2025) [38]	Detecting sophisticated cyberattacks with temporal & spatial dependencies and improving interpretability	Hybrid Seq2Seq with ConvLSTM; CNN for spatial features, LSTM for sequence learning; XAI (LIME) for interpretability	CIC-IDS2017, CIC-ToN-IoT, UNSW-NB15	Accuracy of 99%, 98%, 98% for CIC-IDS2017, CIC-ToN-IoT, UNSW-NB15 datasets respectively
Alsubaei et al. (2024) [39]	Real-time phishing attack detection with efficient computation	EARN (autoencoder + ResNet) feature extraction + ResNeXt-GRU classifier; SMOTE for data balancing; Jaya optimization	Real phishing datasets	Accuracy: 98%, outperforming SOTA by 11–19%; low false positives/negatives; avg. runtime 36.99s
Shahid et al. (2024) [40]	IoT RPL-based network attack detection (Black Hole, Flooding, DODAG Version Number, Reduced Rank)	Transformer-based deep learning model compared with ML baselines (RF best traditional)	ROUT-4-2023	Random Forest: 99% accuracy; Transformer: 98%, F1-score 97%, training time 16.8 min (5 iterations)
Abuali et al. (2023) [41]	Multiclass intrusion detection in social media networks	Hybrid SVM + deep learning fusion	CSE-CIC-IDS2018	Accuracy, Precision, Recall, Specificity, F1-score: 100%; 0% false positives
Zahid & Bharati (2025) [42]	IoT cyberattack detection with high-dimensional and temporal dependency learning	Hybrid CNN + BiLSTM with LASSO feature selection and SMOTE balancing	KDDCup99, NSL-KDD, CIC-IDS2017	Accuracy: 99.9%, 99.8%, and 98.0% respectively; improved precision, recall, F1-score

Table 15 Hybrid Learning Approach: Contribution vs Limitation

Author (Year)	Key Contribution	Limitation
Hariharan et al. (2025) [38]	Hybrid spatial-temporal IDS model with XAI integration (LIME) for interpretability	No analysis of deployment constraints (computational overhead, memory, latency)
Alsubaei et al. (2024) [39]	Unified feature extraction, balancing, and deep learning pipeline with high phishing detection accuracy	No evaluation on resource-constrained devices, no adversarial robustness, no explainability
Shahid et al. (2024) [40]	New ROUT-4-2023 dataset & transformer model for IoT RPL attack detection	No resource-constraint analysis, no interpretability, no adversarial robustness
Abuali et al. (2023) [41]	SVM + deep learning fusion achieving perfect metrics on social media network IDS	Possible overfitting (single dataset), no scalability, no interpretability
Zahid & Bharati (2025) [42]	CNN-BiLSTM hybrid with feature selection and balancing for robust IoT attack detection	No deployment feasibility study, no interpretability for low-resource IoT

8 Explainable AI Techniques

Recent machine learning-based IDS research has been directed towards enhancing explainability. Neural Attention Models [43], Shapley Additive explanations (SHAP) [44], and Local Interpretable Model-agnostic Explanations (LIME) are a few of the methods that have been increasingly utilized. A significant study [45] utilized XAI methods for key feature selection, reducing runtime and enhancing accuracy while providing greater insight into the correlation between features and attacks. Yet, to fully comprehend intrusion techniques, the nature of an IDS must be interpretable. GASPARET et al., (2024) [46] solves the issue of explainability for machine learning-based Intrusion Detection Systems (IDS), particularly IoT environments where model interpretability plays an indispensable role to take informed cybersecurity measures. While emerging IDS models may exhibit superior performance, their black-box design hinders interpretability. To this end, the authors apply two of the most widely used

XAI techniques LIME and SHAP to a black-box IDS model to render intrusion detection more interpretable. The work suggests a perturbation-based validation approach that verifies whether replacing the most significant features (identified by the XAI methods) would alter the model’s output, having an idea of model sensitivity and feature significance. LIME and SHAP are evaluated both numerically and through a survey to quantify interpretability improvements. The primary contribution lies in demonstrating how explainability techniques enhance IDS model transparency and benefit cybersecurity professionals in terms of diagnosis and continuous improvement. However, the research does not examine detection accuracy, real-time efficiency, or system-level deployment, though.

An Interpretable Generalization Mechanism (IG) is introduced by Hao-Ting Pai et al., (2024) [47] to enhance both accuracy and interpretability in Intrusion Detection Systems (IDS). Contrary to the typical black-box IDS approaches, IG trains interpretable patterns from network flows to distinguish between normal and abnormal behavior, promoting interpretability and aiding cybersecurity forensics. It is compared on three benchmarking datasets NSL-KIDD, UNSW-NB15, and UKM-IDS20 across various training-to-test ratios (10%–90%), and its excellent performance is documented (e.g., AUC of up to 0.99 and Recall of up to 1.0 on UNSW-NB15 and UKM-IDS20). Its key strength is that it can detect novel anomalies without pre-exposure with strong generalization power. IG’s explainable decision-making provides actionable feedback on intrusion channels, which is crucial for threat analysis and forensic analysis. But the paper does not address computational effectiveness, high-throughput setting scalability, or integration problems with real-time systems. Despite these limitations, IG is a valuable contribution to the design of XAI-guided IDS in that it is enhancing detection performance while providing transparent, evidence-based explanations.

SHTAYAT et al., (2023) [48] proposes an interpretable ensemble deep learning-based Intrusion Detection System (IDS) for Industrial Internet of Things (IIoT) network security. To address the common limitations of deep learning IDS models i.e., high false positives and non-interpretable decision-making the framework includes Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to provide model prediction transparency. The proposed ensemble technique, experimented with using the ToN.IoT dataset, is demonstrated to achieve higher detection accuracy and robustness compared to a baseline extreme learning machine (ELM) classifier and other conventional approaches. The key contribution is the combination of ensemble deep learning for improved detection performance and explainability techniques for helping cybersecurity analysts interpret and validate IDS decisions. However, the research does not address computational overhead, real-time IIoT deployment latency, or the system’s robustness to adversarial conditions specifically. Despite these limitations, it moves IDS research a step ahead by demonstrating that it is possible to integrate explainability into high-performing ensemble models and utilize it for building trust and facilitating real-world deployment in critical IIoT infrastructure.

OSVALDO ARRECHE et al., (2024) [49] answers the call for AI-based Intrusion Detection Systems (IDS) explainability by suggesting an end-to-end method of comparing black-box Explainable AI (XAI) techniques in the IDS domain. The method focuses on two prevalent XAI techniques SHAP and LIME and tests their

performance using six benchmark measures: descriptive accuracy, sparsity, stability, efficiency, robustness, and completeness. These measures adopt both AI explainability traits and IDS domain-specific needs. The system is tested on three popular intrusion detection datasets and seven AI models of varying architectures, and the comprehensive assessment of the strengths and limitations of current black-box XAI methods in network security is reported. The contribution is made mainly by bridging a gap between cybersecurity practitioners and XAI research through providing a baseline evaluation toolset and making its source code publicly available. However, the work primarily focuses on explanation quality at the cost of improving detection performance or deployment feasibility testing within real-time environments. Despite that limitation, the research greatly advances and benchmarks XAI methods for application in IDS contexts.

Abou et al., (2022) [50] addresses the lack of interpretability in deep learning-based Intrusion Detection Systems (IDS) for IoT networks, as a result of which cybersecurity practitioners are withholding adoption and trust. In this research, the authors present a novel framework that integrates Explainable Artificial Intelligence (XAI) techniques RuleFit, LIME, and SHAP into a novel deep neural network (DNN)-based IDS. The framework offers both global explanations (feature importance) and local explanations (for individual predictions), making it more transparent and easier for cybersecurity practitioners to understand and validate IDS decisions. Experimental confirmation on NSL-KDD and UNSW-NB15 datasets demonstrates that the proposed framework maintains high detection performance while significantly improving interpretability. The main contribution lies in bridging the gap between high-performance deep learning detection models and practical interpretability solutions for IoT security. However, the paper does not mention computational overhead, scalability to real-time IoT deployments, or robustness to adversarial situations. Despite these limitations, this study advances XAI-enabled cybersecurity by introducing an explainable, expert-friendly IDS system for IoT environments.

An explainable feature selection algorithm for encrypted network traffic intrusion detection was proposed by Xuejiao Chen et al., (2024) [51] to addresses the two-folded challenge of model complexity and unexplainability of deep learning-based Network Intrusion Detection Systems (NIDS). The method employs SHAP-based explanation combined with causal analysis to identify and select significant features to achieve feature dimension reduction with guaranteed detection reliability. Experiments conducted on two well-known datasets, CICIDS2017 and NSL-KDD, using Convolutional Neural Networks (CNN) and Random Forest (RF) classifiers provide improved detection capacity with reduced computational complexity compared to baseline models. The major contribution lies in bridging explainable AI techniques to feature optimization to facilitate the development of lightweight yet interpretable NIDS models. However, the study does not discuss deployment factors such as real-time inference, memory constraints, or adversarial network traffic robustness. Despite these limitations, this work offers a useful contribution by introducing a feature selection methodology that enhances efficiency as well as interpretability for intrusion detection based on DL.

Explainable AI (XAI) holds exceptional promise for African cybersecurity teams by improving trust, auditability, and analyst understanding—key for contexts with

Table 16 Summary of Explainable AI Techniques used in the study

Author (Year)	Problem Addressed	XAI tech- niques	Dataset	Performance
Gaspar et al. (2024) [46]	Lack of explainability in ML-based IDS for IoT environments	Black-box IDS interpreted with SHAP & LIME; perturbation-based validation for sensitivity analysis	Not specified	Improved interpretability and transparency, no detection accuracy reported
Hao-Ting Pai et al. (2024) [47]	Improve accuracy and interpretability in IDS; generalization for novel anomalies	Interpretable Generalization Mechanism (IG)	NSL-KDD, UNSW-NB15, UKM-IDS20	AUC up to 0.99, Recall up to 1.0, enhanced model explainability
Shtayat et al. (2023) [48]	High false positives & non-interpretable DL IDS for IIoT	Interpretable ensemble DL IDS using SHAP & LIME	ToN_IoT	Increased transparency alongside detection accuracy
Osvaldo Arreche et al. (2024) [49]	Compare and benchmark XAI techniques for IDS (SHAP vs LIME)	End-to-end evaluation framework for six explainability quality metrics	Three IDS datasets, seven AI models	Baseline benchmarking; explanation quality assessment, not detection accuracy
Abou et al. (2022) [50]	Lack of interpretability in DL-based IoT IDS	XAI-enabled DNN-based IDS with RuleFit, LIME, SHAP (global & local explanation)	NSL-KDD, UNSW-NB15	Enabled multi-level interpretability
Xuejiao Chen et al. (2024) [51]	Model complexity & unexplainability in encrypted traffic IDS	SHAP + causal analysis for feature selection; CNN & RF classifiers	CICIDS2017, NSL-KDD	Enhance explainability with reduced computational complexity

Table 17 The Contribution and Limitation of Explainable AI Techniques in Intrusion Detection

Author (Year)	Key Contribution	Limitation
Gaspar et al. (2024) [46]	Demonstrated SHAP & LIME improve IDS interpretability; perturbation-based sensitivity validation	No analysis of on the trade-of on detection accuracy, computational efficiency
Hao-Ting Pai et al. (2024) [47]	Introduced Interpretable Generalization (IG) mechanism for novel anomaly detection with interpretable patterns	No discussion on computational efficiency or on real-time deployment
Shtayat et al. (2023) [48]	Integrated SHAP & LIME with ensemble DL for interpretable IIoT IDS	Lacked discussion on the overhead analysis, real-time latency, or adversarial robustness
Osvaldo Arreche et al. (2024) [49]	Benchmarked SHAP and LIME with six quality metrics across multiple datasets and models; public toolset provided	Focused on explanation quality only, ignored detection performance and deployment feasibility
Abou et al. (2022) [50]	Provided explainable IoT IDS using RuleFit, LIME, SHAP with high interpretability	No scalability, limited overhead, or adversarial robustness analysis
Xuejiao Chen et al. (2024) [51]	Proposed SHAP-based feature selection for interpretable, efficient IDS	No real-time inference or resource-constraint deployment analysis

limited automation expertise. Integrating interpretable and low-cost AI models could enable local CERTs and public agencies to adopt AI defense systems without relying on external black-box solutions.

9 Cybersecurity Threats and Attacks in Africa

Africa’s accelerating digital transformation driven by mobile finance, e-governance, renewable energy, and ICT expansion—has significantly increased exposure to cyber threats. Despite technological progress, most African nations continue to face limitations in cybersecurity infrastructure, outdated tools, insufficient datasets, and low public awareness. These challenges have made the continent highly susceptible to ransomware, phishing, data breaches, and financial fraud. Recent empirical and technical studies across sectors consistently reveal that inadequate readiness, weak governance, and limited capacity hinder the development of resilient, AI-enabled defense systems.

Omollo and Maharaj (2023) [52] examined cybersecurity behavior among users of Kenya’s mobile financial services and found that perceived threats, low awareness, and high protection costs negatively impact adherence to security practices. Similarly, Kariuki et al. (2023) [53] studied small-scale traders in Southern Africa and reported frequent mobile-based hacking incidents resulting from poor digital literacy and insufficient protection mechanisms. Both studies highlight the heightened vulnerability of informal sector participants and entrepreneurs in Africa’s expanding digital economy.

In the education sector, Eltahir and Ahmed (2023) [54] reported low cybersecurity awareness among Sudanese university students, citing limited institutional support and a lack of formal training. Mitrovic et al. (2025) [55] extended this perspective in South African TVET colleges, proposing a framework for cultivating cybersecurity culture and awareness. These findings collectively emphasize that human factors—particularly digital literacy and behavioral resilience—remain central to strengthening Africa’s cyber defense.

At a continental level, Tahiru (2024) [56] linked the rapid expansion of ICT and IoT infrastructure to increasing cyberattack exposure, noting that policy and regulatory responses often lag technological growth. Beyene et al. (2024) [57] demonstrated that national cybersecurity readiness in Sub-Saharan Africa depends strongly on digital infrastructure and governance maturity, identifying economic and institutional variables as key determinants of resilience. Pashentsev (2024) [58] examined psychological and AI-driven threats within BRICS nations, including South Africa, and called for ethical governance of AI-enabled systems. Ndlovu and Tsibolane (2024) [59] reinforced this viewpoint by reviewing AI-based cyber defense architectures suited to Africa’s low-resource environments, proposing explainable, adaptive, and human-in-the-loop security models to enhance transparency and trust.

Continental threat intelligence further validates these findings. INTERPOL (2025) [17] identified cyber-enabled crimes as a medium-to-high share of total offenses across African member states, with ransomware, business email compromise (BEC), and online scams being the most common. Weak cross-border coordination and inconsistent incident reporting were found to exacerbate these vulnerabilities. Similarly, Kaspersky (2025) [18] observed rising spyware and credential theft across African enterprises, largely due to outdated infrastructure and the lack of Security Operations Centers (SOCs). In the renewable energy domain, Venkatachary and Alagappan (2025) [19] emphasized fragile SCADA systems and minimal cybersecurity investment as key threats to Africa’s energy transition.

From a technical standpoint, several studies have highlighted the potential of AI and edge computing in addressing these challenges. Semou Ndiaye et al. (2024) [60] surveyed cybersecurity in critical infrastructures across developing countries and advocated for AI-driven intrusion detection using Machine Learning (ML) and Deep Learning (DL) techniques to overcome resource limitations. These findings reveal that Africa’s cybersecurity challenge extends beyond technology to include policy gaps, data unavailability, and limited AI literacy. Future research should emphasize developing lightweight, explainable, and trustworthy AI systems specifically optimized for African digital ecosystems. Collaboration among universities, government agencies, and telecom operators is essential for creating open, region-specific datasets and

fostering capacity building. Norman Nelufule (2024) [61] explored human factors contributing to security breaches and argued that organizational culture and awareness are as critical as technical measures. Mbuguah and Gichuki (2024) [62] examined how emerging technologies—AI, Blockchain, and Edge Computing—can advance digital resilience but cautioned that Africa’s cybersecurity readiness remains low. Likewise, Masike Malatji (2024) [63] analyzed 5G/6G vulnerabilities, including adversarial AI and quantum attacks, urging African nations to develop adaptive, context-aware defenses.

In applied AI contexts, Ogore et al. (2024) [64] presented an offline Edge AI model for waterborne disease detection in Rwanda, showcasing how localized, energy-efficient AI systems can perform real-time inference without cloud connectivity—an approach that can be adapted for cybersecurity in rural or bandwidth-limited regions. Similarly, Afachao and Abu-Mahfouz (2024) [65] proposed reinforcement learning-based edge computing models that optimize energy and latency, offering design insights for lightweight, deployable AI systems in low-resource environments.

Overall, the reviewed literature and reports affirm that Africa’s cyber threat landscape is multidimensional—driven by rapid digitization, institutional weaknesses, and uneven technical preparedness. Addressing these issues requires context-specific, explainable, and computationally efficient AI-driven cybersecurity solutions designed for Africa’s economic and infrastructural realities. Building resilience will depend not only on algorithmic innovation but also on sustained investment in digital literacy, capacity building, and cross-border policy coordination to ensure inclusive, AI-enabled security and empowerment across the continent.

10 Performance Metrics

To determine the efficacy of the deep learning model, performance analysis with the help of performance metrics is to be utilized. Through this systematic review, we identified the widely used performance metrics, such as accuracy, sensitivity, specificity, precision, F1-score and AUC-ROC values. These performance metrics are computed depending on true positive (TP), true negative (TN), false positive (FP) and false negative (FN) values, where TP (True Positive) is the number of positive cases identified correctly, TN (True Negative) is the true negative rate, which indicates the number of negative cases correctly identified, FP (False Positive) is the false positive rate that determines the number of instances classified as positive, but are actually negative and FN (False Negative) is the number of positive cases incorrectly reported as negative.

Accuracy:

Accuracy in a classification problem is the ratio of the number of correct predictions to the total number of predictions made by the deep learning model. It is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Table 18 Key Contributions and Limitations of African Cybersecurity Studies (2023–2025)

Author (Year)	Key Contribution	Limitation
Omollo & Maharaj (2023) [52]	Applied behavioral modeling to analyze cybersecurity practices in Kenya’s mobile finance ecosystem.	Limited to user perception; lacks evaluation of technical defense mechanisms.
Kariuki et al. (2023) [53]	Highlighted vulnerabilities among informal traders in digital marketplaces.	Qualitative focus; lacks large-scale empirical validation.
Eltahir & Ahmed (2023) [54]	Quantified cybersecurity awareness in higher education institutions.	Restricted to Sudan; lacks comparative regional insight.
Tahiru (2024) [56]	Provided a continental overview of ICT-driven cyber risk evolution.	Theoretical; lacks quantitative data or sectoral depth.
Venkatachary & Alagappan (2025) [19]	Bridged technical and governance perspectives for securing renewable energy systems.	Conceptual; yet to be empirically validated across African grids.

Sensitivity (Recall)

: Sensitivity calculates the ratio of true positive instances that are accurately recognized. It is also called recall in general.

$$Sensitivity(Recall) = \frac{TP}{TP + FN} \quad (2)$$

Specificity:

Specificity measures the proportion of true negative cases that are correctly identified.

$$Specificity = \frac{TN}{TN + FP} \quad (3)$$

Precision:

It is the proportion of positive identifications that are actually correct:

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

F1-Score:

F1-score is the harmonic mean of recall and precision. It weights the trade-off between them:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

AUC-ROC:

ROC is the abbreviation for Receiver Operating Characteristics. The ROC curve is a plot of Recall (Sensitivity) on the y-axis against the False Positive Rate (FPR), given by $(1 - \text{Specificity})$, on the x-axis. The Area Under the ROC Curve (AUC) is a measure of the model's general capability in discriminating between the classes.

11 Discussion

This systematic review demonstrates that Artificial Intelligence, in the form of machine and deep learning, is playing a leading role in improving cyber attack prediction systems in a variety of domains, such as IoT networks, smart grids, financial services, and corporate infrastructures. Backed by the performance trends and model comparisons presented in the review figures ?? and ??, it can be seen that ensemble learning and hybrid architectures are dominating state-of-the-art accuracy rates, with the majority of the top-performing models achieving over 98–99% detection accuracy. A frequent conclusion across the analyses reviewed is that no single model or approach dominates all others on all metrics and on all datasets. Rather, successful solutions adopt domain-specific optimization techniques: e.g., light CNNs optimize for edge computing [28], but GRU- or LSTM-based models optimize for temporal sequence analysis in smart infrastructure [29, 38]. This specialty is critical to the achievement of an effective trade-off between detection quality and computational manageability, especially in real-time or resource-constrained usage.

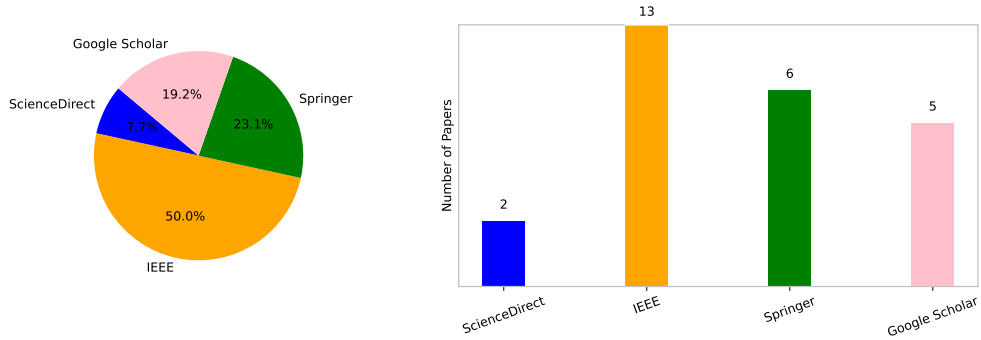


Fig. 2 Reviewed papers distribution based on searching database

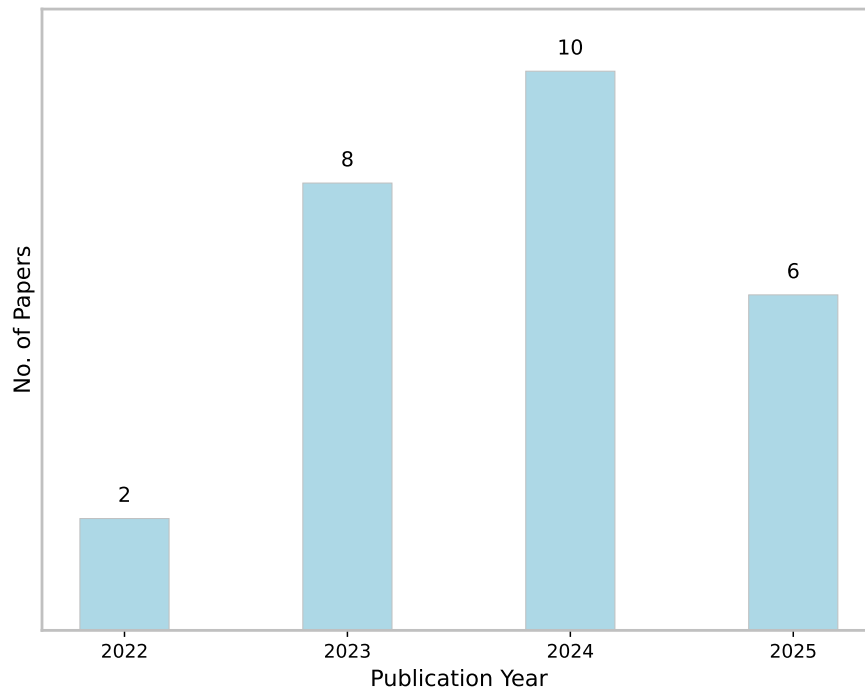


Fig. 3 Yearly distribution of reviewed papers

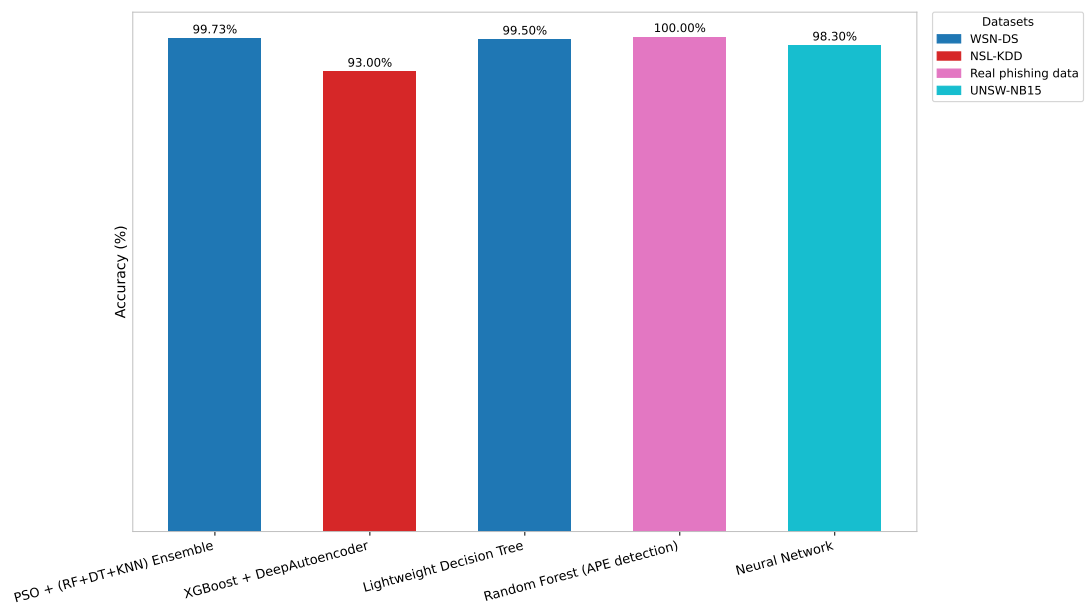


Fig. 4 The performance of various Machine Learning Models

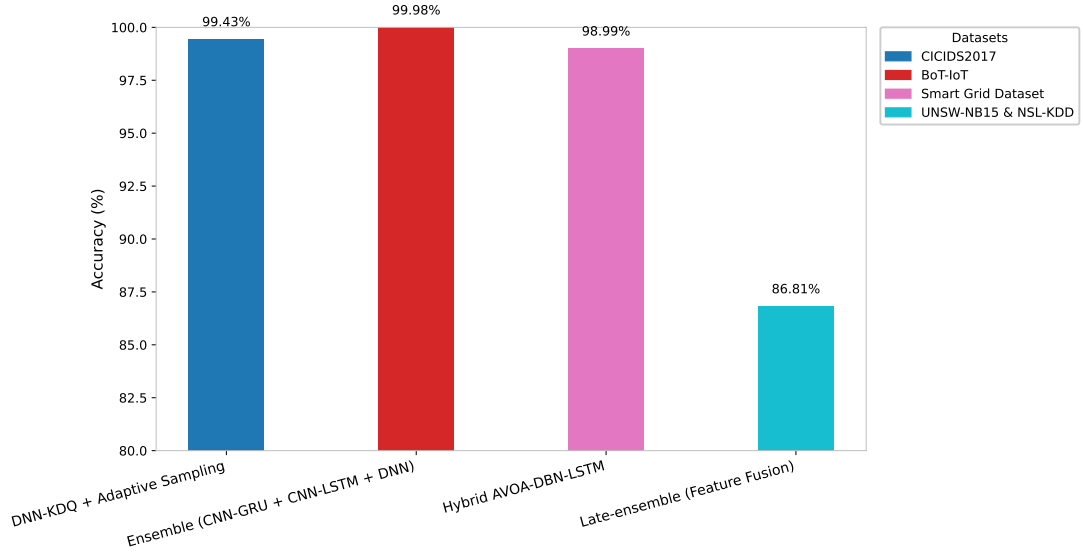


Fig. 5 The performance of various Deep Learning Models

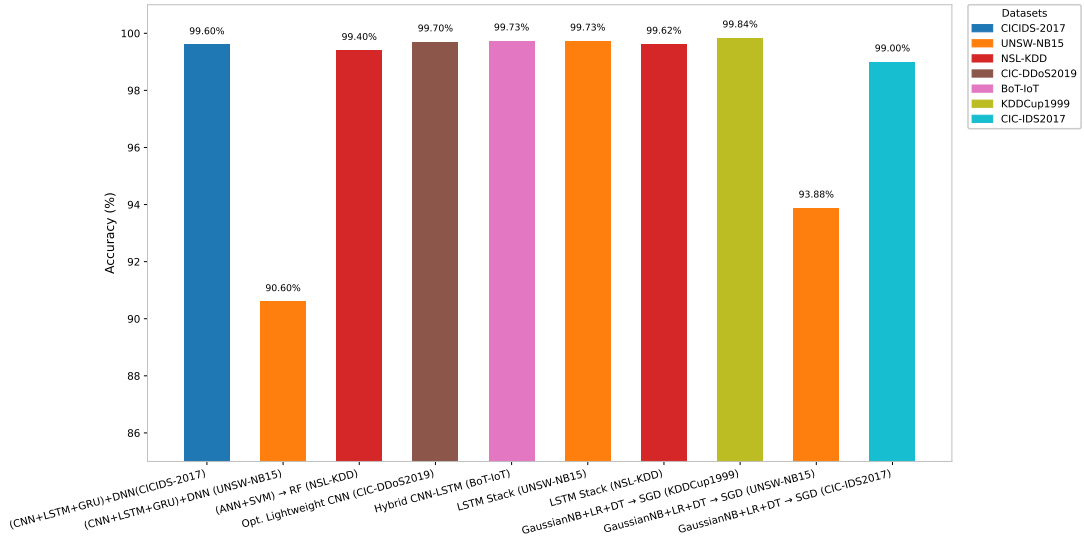


Fig. 6 The performance of various Ensemble Learning Approaches

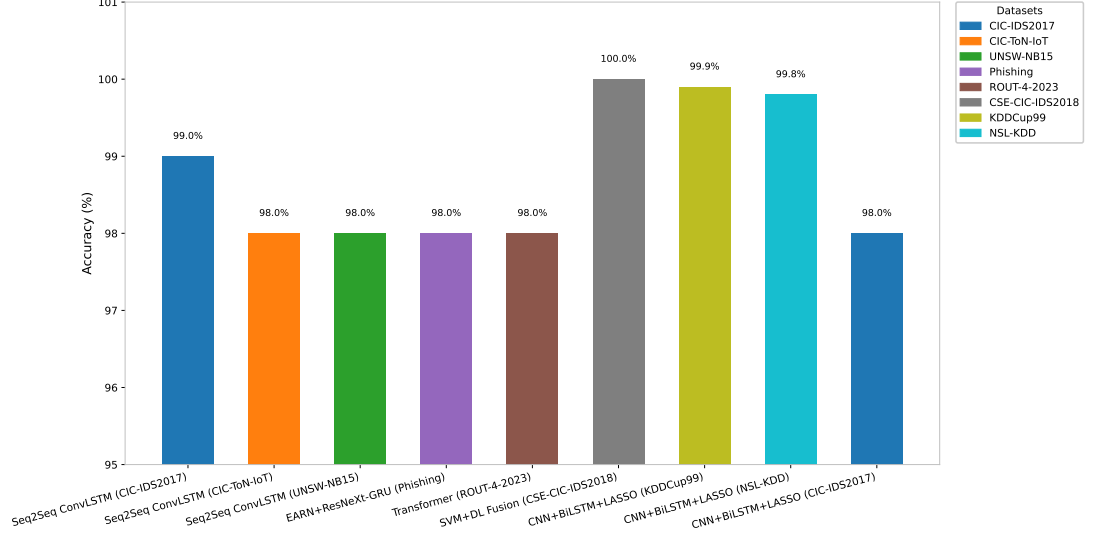


Fig. 7 The performance of various Explainable Deep Learning Models

A second top trend is strategic ensembling of several algorithms. Ensemble and hybrid models using stacking, majority voting, and late fusion usually outperform single-algorithm baselines [32–35, 37, 38, 41, 42]. These architectures offer better robustness to noisy or imbalanced data and demonstrate better generalization on multiclass classification tasks. However, they also come with higher model complexity, longer training times, and increased energy consumption, which are drawbacks to real-world deployment. The growing adoption of Explainable AI (XAI) techniques across cybersecurity models also supports a growing necessity for explainability and transparency. Although black-box models carry out extremely high accuracy, they do not necessarily provide insight into decision-making, limiting their use in critical infrastructure and security operations centers. XAI systems based on SHAP, LIME, and RuleFit [46, 48, 50, 51] offer increased analyst trust and accountability at times but sometimes at the cost of lower detection performance or computational burden. The balance between model interpretability and prediction performance is an open research topic. Moreover, the reliance on benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15 can be seen in most of the research. While the adoption of these datasets presents common platforms for evaluation, their excessive utilization diminishes understanding of model performance under dynamic or adversarial real-world conditions. Few experiments test their models against real network traffic, application-specific datasets (e.g., WSN-DS, DAPT2020), or adversarially perturbed samples—indicated by a lack of deployment preparedness. However, the majority of benchmark datasets originate from non-African network environments and fail to capture local traffic behaviors or attack typologies seen across African enterprises and government systems. This reliance on external datasets limits model generalization and hinders the development of regionally trustworthy AI-based defense mechanisms. On the whole, the

literature under review indicates that AI-centered cybersecurity research is on its way to becoming mature because high technical performance is being achieved under controlled conditions. Adding figure-supported results also demonstrates fragmentation of criteria for evaluation, lack of common metrics for XAI, and limited deployment testing. Bridging the experimental success interpretability effectiveness gap needs future research to concentrate on real-time scalability, energy-effective architectures, adversarial robustness, and federated or privacy-preserving learning models. These findings underscore the need for localized research efforts aimed at creating representative African cybersecurity datasets and lightweight AI models deployable on limited infrastructure. Moreover, regional collaboration among universities, telecom operators, and national CERTs can help bridge the data and capacity gaps, enabling more inclusive and sustainable cybersecurity resilience across the continent.

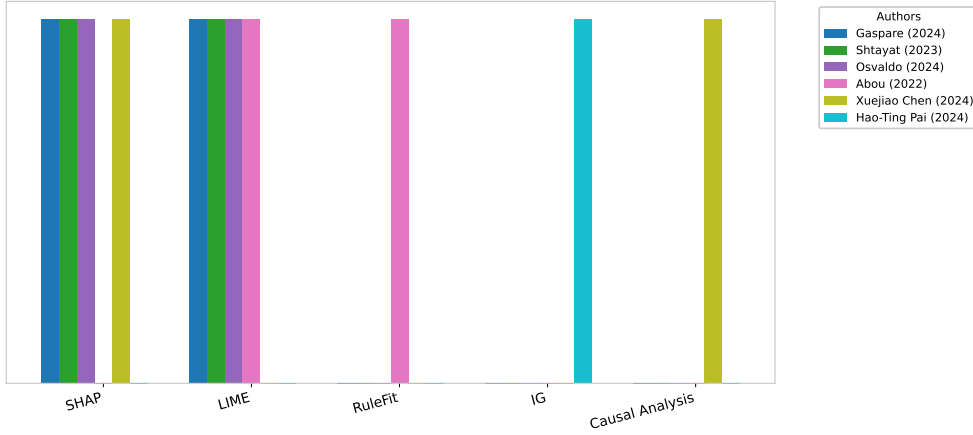


Fig. 8 The various Explainability techniques used in Cybesecurity application

12 Challenges and Future Direction

Although significant progress has been made in AI-based cybersecurity, several open challenges still hinder its practical applicability and reliability. One persistent issue is the over-reliance on standard benchmark datasets such as NSL-KDD, CICIDS2017, and UNSW-NB15, which do not adequately represent the diversity of real-world or region-specific cyber threats. This limitation is especially evident in the African context, where the lack of localized datasets reflecting regional network traffic and attack vectors constrains the development of context-aware defense models. Furthermore, most proposed systems have not been validated under real-time or resource-limited environments typical of African infrastructures, such as IoT, mobile, and edge networks. While ensemble and hybrid models offer improved accuracy, their computational intensity leads to longer training times, higher energy consumption, and limited scalability—factors that challenge deployment in low-resource settings. Explainable AI (XAI) techniques like SHAP and LIME are gaining traction, yet their broader adoption in Africa is restricted by additional computational overhead, limited

expertise, and the absence of benchmark evaluation frameworks. Adversarial robustness, latency, and cost-effective integration also remain underexplored, particularly in developing regions where cybersecurity budgets and technical capacity are limited. Therefore, future research should emphasize the design of lightweight, energy-efficient, and adversarially robust AI architectures tailored for Africa's bandwidth and power-constrained environments. Additionally, there is a critical need for the creation of multi-faceted, application-oriented datasets that incorporate African network characteristics and socio-technical realities. Building combined frameworks that balance high detection accuracy with interpretability will be essential for empowering analysts and institutions across the continent to make timely, data-driven security decision.

13 Conclusion

In conclusion, this systematic literature review reaffirms the reality that Artificial Intelligence specifically via machine learning, deep learning, and explainable architectures is presently a fundamental cornerstone to the augmentation of cyber attack forecasting and intrusion detection. The reviewed studies point to the fact that no single model architecture excels on all dimensions or domains but rather that performance is inextricably linked to domain-specific adaptation, dataset suitability, and computationally induced constraints. Hybrid and ensemble models continue to outperform traditional approaches, and explainable AI brings in the much-desired transparency and analyst confidence, especially in finance and critical infrastructure domains. Real-world deployment is still held back by adversarial robustness, real-time responsiveness, and model interpretability in production environments. Closing the experimental efficacy-deployment viability gap, therefore, is required to advance AI-powered cybersecurity systems further. In the future, solutions that include robust detection, minimal resource consumption, scalability, and transparency will be instrumental in safeguarding against ever-growing sophisticated and common cyber attacks in centralized and distributed digital systems. To conclude, this review bridges global AI advancements with Africa's cybersecurity realities. By prioritizing explainability, energy-efficient architectures, and regional collaboration, it outlines a practical pathway toward building resilient and inclusive AI-powered defense systems for Africa's emerging digital economy.

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